Autonomous Search and Multi-Objective Assessment of Theoretical Wormhole Geometries

Revised English preliminary research report | 2 November 2026 | Documented run: 30 October 2026 | C++17 and paceval SDK 4.25 | Not independently peer-reviewed

Research status. A computational experiment in classical general relativity. This report is not an astronomical detection, a physically realizable construction, or a claim of global optimality. Numerical TINC checks were performed at discrete radial coordinates, not over a continuous domain.

Plain-language summary

The application automatically composes mathematical descriptions of hypothetical static wormholes and compares them using two competing numerical criteria: a radial measure of null-energy-condition (NEC) violation and sampled static tidal curvature. For the selected candidate, the first measure is approximately 89.5% below the specific massless Ellis reference under the same fitness protocol. This does not establish an 89.5% reduction in physically required exotic matter. The candidate has greater static tidal curvature; whether a traveler would experience larger acceleration depends on physical throat scale, motion and reference frame.

Abstract

Method. A grammar-constrained evolutionary search constructs expression trees for the Morris–Thorne shape and redshift functions. The C++ application performs symbolic differentiation and optimization; paceval evaluates prepared expression handles through scalar and batch calls. A subset of candidates is inspected using two objectives, followed by finite pointwise TINC checks of the selected solution.

Results. Seed 20261029, 100 new candidate evaluations; 24 fine-grid Pareto-preview candidates, 19 nondominated within that subset. Selected-candidate negative NEC fitness 0.10353962119525634 compared with Ellis 0.99001188178969057. A separate diagnostic quadrature reports 0.10356062635634017 and 0.99002294791025836. Maximum sampled static radial tidal component: 1.9935043450750545; maximum Kretschmann scalar: 27.896207978563247. These are sampled, normalized diagnostics.

1. Geometry and generated expression

Using geometrized units (G = c = 1) and a normalized throat radius r0 = 1, the metric is

$$ds^2 = -\exp(2\Phi(r))\,dt^2 + dr^2 / [1 - b(r)/r] + r^2\,(d\theta^2 + \sin^2\theta\,d\varphi^2).$$

The following compact expression is algebraically equivalent to the output recorded in wormhole_final_candidate.txt. The source file records a = 1.7573959995777177; this precision is retained rather than the slightly different final digits in earlier manuscript drafts.

$$q(r) = (r - 1)/r;\ a = 1.7573959995777177;\ \gamma = 1.9935043450750545$$

$$U(q) = q + q^3 / [(1 + q^2)(1 + \cos^2 a)]$$

$$b(r) = [1 + q(r)\,U(q(r))]/r;\ \Phi(r) = -\gamma/r$$

At the throat b(1) = 1 and b′(1) = −1, fulfilling the local flare-out condition. Within the program-defined expression grammar the exterior asymptotic form follows structurally, but the sampled checks do not prove globally stable traversability.

Reproducible text expressions

```
q=(r-1)/r
U=q+(q*q*q)/((1+q*q)*(1+cos(a)*cos(a)))
a=1.7573959995777177
gamma=1.9935043450750545
b=(1+q*U)/r
Phi=(0-gamma)/r
```

2. The exact numerical objective and its interpretation

$$8\pi(\rho + p_r) = b'(r)/r^2 - b(r)/r^3 + 2[1 - b(r)/r]\Phi'(r)/r$$

$$J_NEC = \int_1^{100} (r^2/2)\, \max[0, -8\pi(\rho + p_r)]\, dr$$

The fitness estimator is a sampled numerical approximation to this program-defined, finite-interval integral. It is not automatically an averaged null energy condition on complete null geodesics, the Visser–Kar–Dadhich volume integral quantifier, or a physical mass or energy requirement. The value depends on metric normalization, weighting, sampling and quadrature.

Two distinct evaluation paths in the actual CSV

Quantity	Selected candidate	Massless Ellis
negative_NEC_fitness	0.10353962119525634	0.99001188178969057
negative_NEC_quadrature	0.10356062635634017	0.99002294791025836

The approximately 89.5% decrease is calculated by comparing fitness with fitness, not by mixing the distinct numerical estimates. Neither value is a rigorously interval-certified integral. The two estimates should not be presented as an exact identity.

3. Comparison models and physical trade-off

Model	Fitness J_NEC	Max radial tidal	Max K
Massless Ellis	0.9900118818	0	11.99999
Ellis, $\Phi = -0.75/r$	0.2635584156	0.75	14.24999
Constructed U = 1.90	1.4461487422	0	4.02000
Constructed U = 1.99	1.4677552250	0	4.00020
Selected candidate	0.1035396212	1.9935043451	27.89621

The higher static orthonormal curvature is a competing diagnostic. It is not itself the acceleration experienced by a traveler: velocity, worldline, observer frame and dimensional throat radius must be specified. The selected candidate is not claimed to dominate all other known wormhole families.

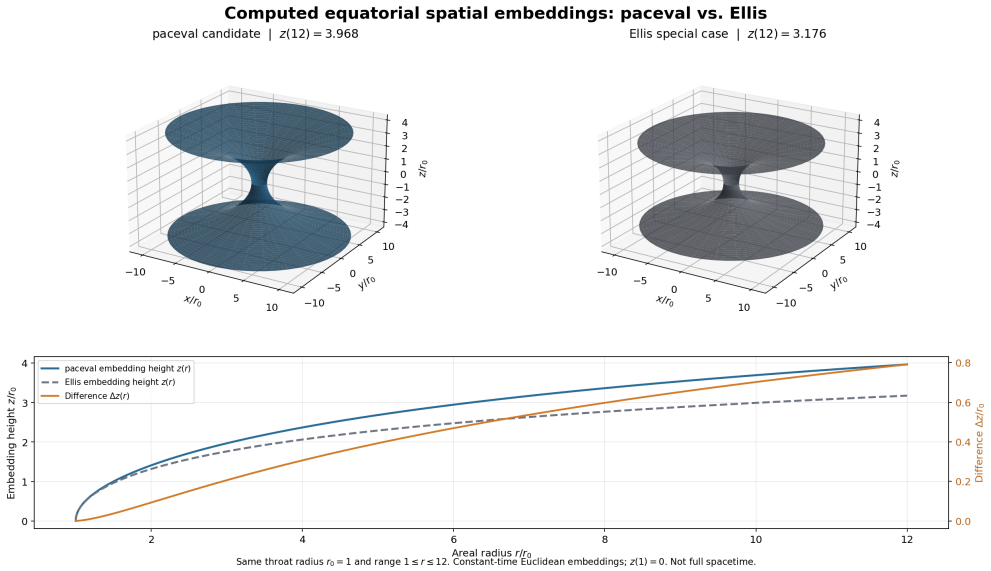


Figure 1. Numerically integrated, same-scale equatorial spatial embeddings for the selected candidate and massless Ellis reference. The difference is defined at matched areal radius, with z(1) = 0. These are spatial slices, not full spacetime visualizations.

4. Pareto exploration

The program studies the numerical objective pair (J_NEC, T_static), where T_static is the largest sampled absolute selected static orthonormal tidal Riemann component. Of 100 evaluated candidates, 24 were included in a fine-grid Pareto preview; 19 were nondominated within those 24, not over a globally exhaustive search.

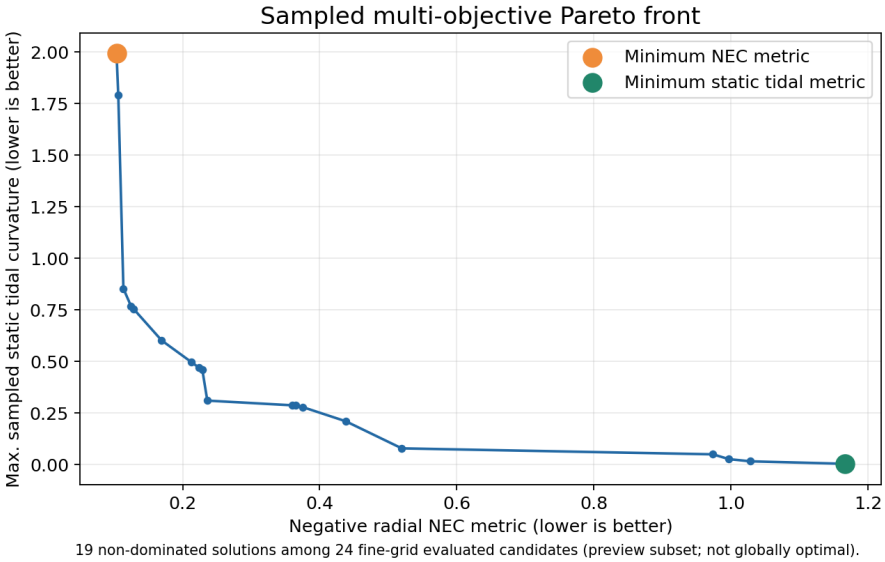


Figure 2. The nineteen nondominated entries from the uploaded wormhole_pareto_front.csv. No global optimality or continuum validation is asserted.

5. Validation, TINC and failure recovery

Three analytic reference geometries were checked against 12 constructed physical fields, including derivative and curvature quantities. The documented selected candidate received 1,024 regular radial TINC evaluations plus 273 adaptive points (17 initial and 256 additional), yielding no unresolved evaluated points. On the regular grid, 142 NEC intervals were classified negative, 882 nonnegative and none inconclusive. A maximum sampled NEC interval width of approximately 5.48×10^{-13} is reported by the archived run.

At one sampled radius, four original expressions produced paceval error code 144. Algebraically equivalent fallback expressions were evaluated for the same point with TINC: four attempts, four recovered, zero unresolved. The operator-level cause of error 144 is not thereby identified or repaired. A successful bound for

finitely many fixed inputs does not bound the function over all intermediate radii; nor does it validate the numerical integral or the optimization search.

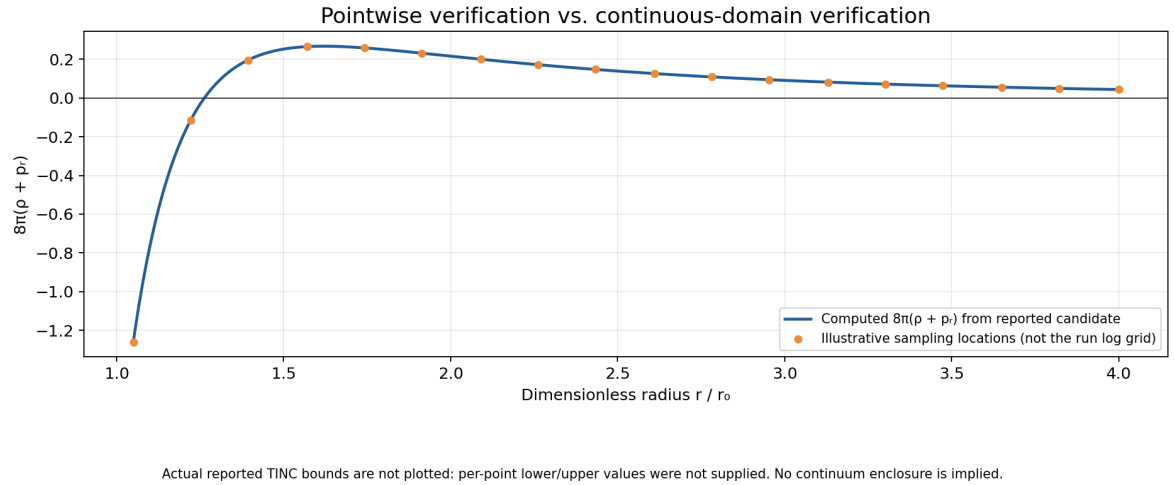


Figure 3. Conceptual visualization of pointwise versus continuous-domain validation. Sampling locations are illustrative, not the original TINC grid; measured interval bounds are not fabricated.

6. Reproducibility and implementation provenance

Actual uploaded execution artifacts: wormhole.cpp; wormhole_discovery_debug.log; wormhole_final_candidate.txt; wormhole_pareto_front.csv; wormhole_reference_comparison.csv. The exact output candidate and Pareto rows originate from these artifacts. The associated figure-generation script, generate_figures.py, is a separate visualization aid and does not implement the autonomous search or TINC.

Setting	Recorded configuration
SDK / language	paceval 4.25 / C++17, Visual Studio
Seed / evaluation budget	20261029 / 100 newly evaluated candidates
Population / elite / max depth	20 / 4 / 5
Search grids	72 coarse / 1,200 fine radial points
Pareto preview	24 of 100 archived candidates
TINC regular / adaptive	1,024 / 273 (17 + 256)
TINC batch size	128 configured (not shown to be optimal)
Selected U parameter	a = 1.7573959995777177

The original source creates the mathematical expressions and their derivatives. Prepared paceval handles are reused and sampled radial computations are batched. The observed timings are environment-specific and are not a CPU-to-FPGA/ASIC performance proof. No independent full rebuild and re-execution of the uploaded program has been performed as part of this manuscript revision.

7. Scientific limitations and prioritized next work

The result is preliminary. Priority work includes: (1) independently reconstruct all Einstein, Ricci and Riemann expressions with explicit sign conventions; (2) assess the full candidate archive, not only the 24-candidate preview, and repeat across independent seeds; (3) obtain rigorous interval-domain bounds and validated quadrature; (4) compare against optimized parametric families under equal evaluation budgets; (5) specify a physical stress-energy source, examine junctions and dynamical stability; (6) evaluate traveler-dependent tidal accelerations after selecting dimensional throat scale; and (7) benchmark performance and energy consumption independently.

8. Conclusion

The successful recorded run supports an exploratory workflow for autonomously generating symbolic candidate geometries and comparing numerical NEC and tidal-curvature objectives. Relative to the selected massless Ellis reference, the candidate markedly decreases the specific sampled NEC fitness objective but increases sampled static tidal curvature. The study does not show an observed wormhole, globally optimal solution, physically reduced exotic-matter requirement, or validated continuum inequality.

References

[1] Morris, M. S.; Thorne, K. S. (1988). Wormholes in spacetime and their use for interstellar travel. American Journal of Physics 56, 395–412. DOI: 10.1119/1.15620.

[2] Visser, M.; Kar, S.; Dadhich, N. (2003). Traversable Wormholes with Arbitrarily Small Energy Condition Violations. Physical Review Letters 90, 201102. DOI: 10.1103/PhysRevLett.90.201102.

[3] Ellis, H. G. (1973). Ether flow through a drainhole: A particle model in general relativity. Journal of Mathematical Physics 14, 104–118. DOI: 10.1063/1.1666161.

[4] Deb, K.; Pratap, A.; Agarwal, S.; Meyarivan, T. (2002). A fast and elitist multiobjective genetic algorithm: NSGA-II. IEEE Transactions on Evolutionary Computation 6, 182–197. DOI: 10.1109/4235.996017.

Provenance: The documented Windows execution in wormhole_discovery_debug.log dated 30 October 2026 is associated with the supplied wormhole.cpp and its original CSV and candidate files. This report updates and extends the earlier six-page English preliminary report and should replace that earlier document, not be represented as an independently peer-reviewed analysis.